

Linear Algebra

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UAB-IDEA

September 2026

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Section 1

Matrices & vectors

What is linear algebra?

Linear algebra studies vectors, matrices, linear equations, and the maps they represent.

Recurring question. When can an equation such as $Ax = b$ be solved, and when is the solution unique?

Why this chapter matters.

- ◆ The natural setting is $\mathbb{R}^n$.
- ◆ Differentiation uses vectors and matrices.
- ◆ Optimization uses rank and invertibility.
- ◆ Statistics uses projections and full-rank conditions.

From equations to a linear system

Model equations. Let income be Y , consumption be C , and exogenous investment be $\bar{I} = 50$.

$$Y = C + \bar{I}, \quad \bar{I} = 50,$$

$$C = 100 + 0.8Y.$$

Move every unknown to the left:

$$Y - C = 50,$$

$$-0.8Y + C = 100.$$

Align the unknowns. Put Y first and C second. The coefficient rows are

$$[1 \quad -1], \quad [-0.8 \quad 1].$$

Fixed terms go on the right; coefficients multiplying unknowns go into the matrix.

Each equation supplies one **row**; the chosen order Y, C fixes the **column order**.

Matrix form and solution check

Matrix form.

$$\underbrace{\begin{bmatrix} 1 & -1 \\ -0.8 & 1 \end{bmatrix}}_A \underbrace{\begin{bmatrix} Y \\ C \end{bmatrix}}_x = \underbrace{\begin{bmatrix} 50 \\ 100 \end{bmatrix}}_b.$$

Thus the two equations are written as $Ax = b$, and $[A \mid b]$ denotes the **augmented matrix** carrying both.

Solve and verify. From $Y = C + 50$ and $C = 100 + 0.8Y$,

$$C = 100 + 0.8(C + 50)$$

$$\Rightarrow C = 700,$$

$$Y = C + 50 = 750.$$

The two rows of $Ax = b$ check the solution:

$$750 - 700 = 50,$$

$$-0.8(750) + 700 = 100.$$

Changing a fixed term changes b ; changing a coefficient on an unknown changes A .

Video: [MIT OpenCourseWare, *The Geometry of Linear Equations*](#).

Definitions

Matrix. $A_{m \times k}$ has m rows and k columns. The entry a_{ij} sits in row i , column j .

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1k} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mk} \end{bmatrix}.$$

Vector. A vector in $\mathbb{R}^k$ is an ordered k -tuple of real numbers. It may be written as a row or as a column:

$$[v_1, \dots, v_k], \quad \begin{bmatrix} u_1 \\ \vdots \\ u_k \end{bmatrix}.$$

For $x, y \in \mathbb{R}^k$ and $\alpha \in \mathbb{R}$, vector addition and scalar multiplication are componentwise:

$$(x + y)_i = x_i + y_i, \quad (\alpha x)_i = \alpha x_i.$$

Video: [3Blue1Brown, Vectors](#).

Inner product

Definition. For $a, b \in \mathbb{R}^n$,

$$a \cdot b = \sum_{i=1}^n a_i b_i.$$

The output is a scalar (number).

Matrix notation. With a^T the row vector holding the entries of the column vector a ,

$$a \cdot b = a^T b = b^T a,$$

also written $a'b$.

Dimension check. The two vectors must have the same dimension. Otherwise the sum $\sum_i a_i b_i$ is not defined.

Video: 3Blue1Brown, *Dot products and duality*.

Inner product: example

Vectors.

$$v_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}.$$

Calculation.

$$\begin{aligned} v_1 \cdot v_2 &= v_1^T v_2 = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} \\ &= 1 \cdot 2 + 2 \cdot (-1) = 2 - 2 = 0. \end{aligned}$$

Since **their inner product is zero**, they are **orthogonal**.
Only vectors of the **same dimension** can be orthogonal.
The zero vector is orthogonal to every vector of the same dimension.

Strictly speaking, dimension is a property of the vector space to which a vector belongs, not of the vector itself. You will see this distinction in Optimization; here I use “dimension” loosely to mean the number of coordinates of a vector in $\mathbb{R}^n$.

Euclidean norm

Norm. For $x \in \mathbb{R}^n$,

$$\|x\|_2 = \sqrt{x \cdot x}.$$

For $x = (3, 4)^T$,

$$\|x\|_2 = \sqrt{3^2 + 4^2} = 5, \quad \|-2x\|_2 = 10.$$

There are many other norms. Again, you will see them in Optimization.

Video: [Jaisohn Kim, Vector and matrix norms.](#)

Norm and orthogonality

Cauchy-Schwarz and the Pythagorean relation

$$|x \cdot y| \leq \|x\|_2 \|y\|_2.$$

Nonzero vectors satisfying $x \cdot y = 0$ are orthogonal. For such vectors,

$$\|x + y\|_2^2 = \|x\|_2^2 + \|y\|_2^2.$$

Further reading: [Boyd and Vandenberghe, vector norms and inner products.](#)

Hyperplanes and half-spaces

Hyperplane. For $a \in \mathbb{R}^n \setminus \{0\}$ and $b \in \mathbb{R}$,

$$H = \{x \in \mathbb{R}^n : a^T x = b\}$$

is a hyperplane with *normal vector* a .

Example. The line

$$x_1 + 2x_2 = 4$$

has normal vector $a = (1, 2)^T$.

Why a is normal. If $u, v \in H$, then

$$a^T(u - v) = a^T u - a^T v = b - b = 0.$$

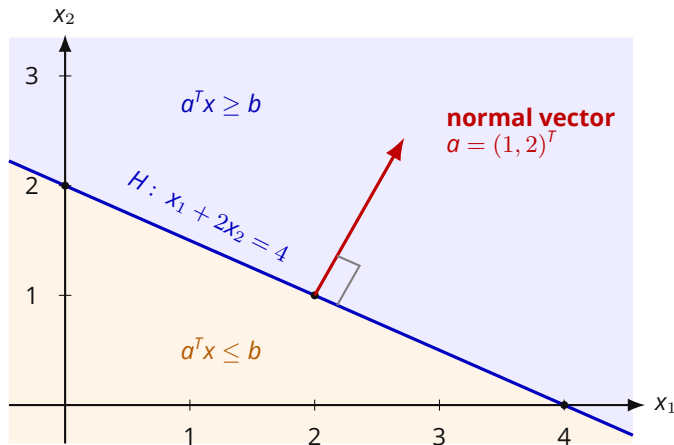
Associated half-spaces.

$$a^T x \leq b, \quad a^T x \geq b.$$

Each inequality includes the boundary H .

Video: [Mathelecs, Hyperplanes and halfspaces.](#)

Hyperplane, half-spaces, and normal vector



The hyperplane is the common boundary of the two half-spaces, and a points in a direction orthogonal to the hyperplane.

Linear combinations and span

Linear combination. Given $v_1, \dots, v_k \in \mathbb{R}^n$, any vector of the form

$$\alpha_1 v_1 + \dots + \alpha_k v_k, \quad \alpha_1, \dots, \alpha_k \in \mathbb{R},$$

is a linear combination of $v_1, \dots, v_k$.

Span.

$$\text{span}\{v_1, \dots, v_k\} = \left\{ \alpha_1 v_1 + \dots + \alpha_k v_k : \alpha_1, \dots, \alpha_k \in \mathbb{R} \right\}.$$

Membership question. A vector w lies in the span if there exist $\alpha_1, \dots, \alpha_k$ such that $\alpha_1 v_1 + \dots + \alpha_k v_k = w$.

Video: 3Blue1Brown, *Linear combinations, span, and basis vectors*.

Linear independence

Linear independence. The vectors $v_1, \dots, v_k$ are *linearly independent* if, for every $\alpha_1, \dots, \alpha_k \in \mathbb{R}$,

$$\alpha_1 v_1 + \dots + \alpha_k v_k = 0 \quad \Rightarrow \quad \alpha_1 = \dots = \alpha_k = 0.$$

That is, only the all-zero choice of coefficients can produce the zero vector.

Dependence. The list is *dependent* if some coefficients, **not all zero**, satisfy

$$\alpha_1 v_1 + \dots + \alpha_k v_k = 0.$$

With $A = [v_1 \ \dots \ v_k]$, the columns are **independent if and only if** the only solution of $Ax = 0$ is $x = 0$.

Uniqueness. Independence makes the coefficients of every vector in the span unique.

Video: [Khan Academy, Introduction to linear independence.](#)

Linear combinations: example

Let

$$v_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \quad w = \begin{bmatrix} 5 \\ 0 \end{bmatrix}.$$

To check whether w is a linear combination of v_1 and v_2 , solve for the weights α and β :

$$\alpha v_1 + \beta v_2 = w \Leftrightarrow \begin{cases} \alpha + 2\beta = 5, \\ 2\alpha - \beta = 0. \end{cases}$$

The second equation gives $\beta = 2\alpha$. Substituting this into the first equation gives

$$\alpha + 2(2\alpha) = 5 \Leftrightarrow 5\alpha = 5 \Leftrightarrow \alpha = 1.$$

Hence $\beta = 2$, and therefore

$$w = v_1 + 2v_2.$$

Linear independence: example

Use the same two vectors

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}.$$

To test whether they are linearly independent, set a linear combination equal to zero:

$$\alpha \mathbf{v}_1 + \beta \mathbf{v}_2 = \mathbf{0} \quad \Leftrightarrow \quad \begin{cases} \alpha + 2\beta = 0, \\ 2\alpha - \beta = 0. \end{cases}$$

The second equation gives $\beta = 2\alpha$. Substituting this into the first equation gives

$$\alpha + 2(2\alpha) = 0 \quad \Leftrightarrow \quad 5\alpha = 0 \quad \Leftrightarrow \quad \alpha = 0.$$

Hence $\beta = 0$. The only solution is $\alpha = \beta = 0$, so $\mathbf{v}_1$ and $\mathbf{v}_2$ are linearly independent.

Subspaces, span, basis & dimension

Subspace. A nonempty set $V \subseteq \mathbb{R}^n$ is a subspace if $u, v \in V$ and $\alpha, \beta \in \mathbb{R}$ imply

$$\alpha u + \beta v \in V.$$

Basis. A basis of a subspace V is a linearly independent list whose span is V .

Span. $\text{span}\{v_1, \dots, v_k\}$ is the set of all linear combinations of $v_1, \dots, v_k$; it is the smallest subspace containing them.

Dimension. $\dim V$ is the number of vectors in any basis of V . In $\mathbb{R}^n$, the standard vectors $e_1, \dots, e_n$ form a basis, so $\dim \mathbb{R}^n = n$.

For the earlier example, $v_1 = (1, 2)^T$ and $v_2 = (2, -1)^T$ are linearly independent in $\mathbb{R}^2$, hence they form a basis of $\mathbb{R}^2$. Video: [MIT 18.06SC, Basis and dimension](#).

Independent pair, dependent triple

Independent pair. From the previous example,

$$\alpha v_1 + \beta v_2 = 0 \Rightarrow \alpha = \beta = 0.$$

Hence (v_1, v_2) is linearly independent.

Dependent triple. Since $w = v_1 + 2v_2$,

$$v_1 + 2v_2 - w = 0.$$

The coefficients $1, 2, -1$ are not all zero.

Dimension bound. Since $\dim \mathbb{R}^n = n$, a linearly independent list in $\mathbb{R}^n$ can contain at most n vectors. Here all vectors lie in $\mathbb{R}^2$, so no three-vector list in this space can be linearly independent.

Section 2

Matrix operations

Addition & scalar multiplication

Entrywise operations. For A, B of the same size and $c \in \mathbb{R}$,

$$(A + B)_{ij} = a_{ij} + b_{ij}, \quad (cA)_{ij} = ca_{ij}.$$

Laws.

$$A + B = B + A,$$

$$A + (B + C) = (A + B) + C.$$

Matrix addition.

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}.$$

Restriction. Addition requires the same dimensions. Scalar multiplication does not change matrix size.

Video: [Khan Academy, Matrix addition and subtraction.](#)

Multiplication

Entry formula. For $A \in \mathbb{R}^{m \times k}$ and $B \in \mathbb{R}^{k \times n}$,

$$(AB)_{ij} = \sum_{\ell=1}^k a_{i\ell} b_{\ell j} = \text{row}_i(A) \cdot \text{col}_j(B).$$

Rules.

- ◆ Inner dimensions must match.
- ◆ Multiplication is associative and distributive.
- ◆ In general $AB \neq BA$, and BA may be undefined.

Numerical calculation. Multiply a 2×2 matrix by a 2×1 vector:

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 10 \end{bmatrix}$$

The result is a 2×1 vector.

Video: 3Blue1Brown, *Matrix multiplication as composition*.

A matrix defines a linear map

Linear map. Let $A \in \mathbb{R}^{m \times n}$. The rule

$$T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m, \quad T_A(x) = Ax,$$

is linear: $T_A(\alpha x + \beta y) = \alpha T_A(x) + \beta T_A(y)$.

Column combination. If the columns of A are $a_1, \dots, a_n$, then

$$Ax = x_1 a_1 + \dots + x_n a_n.$$

Composition. If $B \in \mathbb{R}^{n \times p}$ and $A \in \mathbb{R}^{m \times n}$, then

$$T_{AB} = T_A \circ T_B : \mathbb{R}^p \rightarrow \mathbb{R}^m.$$

Thus $ABx = A(Bx)$: apply B first and A second.

Dimension check. The n columns of A match the n coordinates of its input. Its m rows determine the m output coordinates.

Video: 3Blue1Brown, *Linear transformations and matrices*.

Image, kernel, and rank

Let $A = [a_1 \ \cdots \ a_n] \in \mathbb{R}^{m \times n}$.

Image / column space. The image is the set of all possible outputs:

$$\text{im}(A) = \{Ax : x \in \mathbb{R}^n\} = \text{span}\{a_1, \dots, a_n\}.$$

The rank is the dimension of the image:

$$\text{rank}(A) := \dim \text{im}(A).$$

It counts the independent output directions.

Kernel / null space. The kernel is the set of inputs mapped to zero:

$$\ker(A) = \{x \in \mathbb{R}^n : Ax = 0\}.$$

The rank-nullity theorem gives

$$n = \text{rank}(A) + \dim \ker(A).$$

It describes the input directions mapped to zero.

Solvability. $Ax = b$ is solvable if and only if $b \in \text{im}(A)$.

Independence. The columns of A are independent if and only if $\ker(A) = \{0\}$.

Video: [Let's Get Complex, Kernel, image, and the rank-nullity theorem.](#)

Transpose

Transpose. $(A^T)_{ij} = a_{ji}$: rows and columns are swapped. Whenever operations are defined,

$$(A + B)^T = A^T + B^T, \quad (AB)^T = B^T A^T.$$

Dimension reversal. If A is $m \times n$, then A^T is $n \times m$.

Symmetric. A square matrix is symmetric if $A^T = A$. For any $X \in \mathbb{R}^{m \times n}$, the rules above give

$$(X^T X)^T = X^T (X^T)^T = X^T X, \quad (X X^T)^T = X X^T.$$

Product rule by shape. If A is $m \times k$ and B is $k \times n$, then

$$(AB)^T = B^T A^T$$

has size $n \times m$.

Videos: [MathTheBeautiful](#), [Introduction to the transpose](#); [Transpose of a product](#).

Special matrices

Named square matrices. A square matrix is $n \times n$, and a diagonal matrix has every off-diagonal entry equal to zero. The identity I_n carries ones on the diagonal, the zero matrix carries zeros everywhere, an idempotent matrix satisfies $A^2 = A$, and a symmetric matrix satisfies $S^T = S$.

Examples.

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad 0_{2 \times 2} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix},$$

$$S = \begin{bmatrix} 2 & -1 \\ -1 & 3 \end{bmatrix} \Rightarrow S^T = S.$$

Transpose: example

Start from

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}.$$

Its transpose is

$$A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}.$$

- ◆ Row 1, $[1, 2, 3]$, becomes column 1.
- ◆ The dimensions reverse: $2 \times 3 \mapsto 3 \times 2$.
- ◆ Transposing twice gives back the original matrix:

$$(A^T)^T = A.$$

Choose, calculate, and justify

Let

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}, \quad b = \begin{bmatrix} 3 \\ 6 \end{bmatrix}, \quad c = \begin{bmatrix} 3 \\ 7 \end{bmatrix}.$$

1. Decide whether b and c belong to the span of the columns of A .
2. Compute A^T and decide whether A is symmetric.

Solution and checks

Equations and span. Every vector in the span of A is of the same form: the second row is twice the first. So b is attainable, while c is not attainable.

Transpose.

$$A^T = A,$$

so A is symmetric.

Trace

Definition. For a square matrix $A \in \mathbb{R}^{n \times n}$,

$$\text{Tr}(A) = \sum_{i=1}^n a_{ii}.$$

EXAMPLE

$$C = \begin{bmatrix} 2 & -1 & 4 \\ 0 & 3 & 5 \\ 7 & 1 & -2 \end{bmatrix},$$

$$\text{Tr}(C) = 2 + 3 - 2 = 3.$$

Video: [MEXAMS, Trace: key properties and their proofs.](#)

Trace identities

Linearity and cyclic rotation

$$\text{Tr}(\alpha A + \beta B) = \alpha \text{Tr}(A) + \beta \text{Tr}(B).$$

Compatible factors may be cyclically rotated:

$$\text{Tr}(ABC) = \text{Tr}(BCA) = \text{Tr}(CAB).$$

For compatible dimensions,

$$\text{Tr}(AB) = \text{Tr}(BA).$$

Further reading: [Boyd and Vandenberghe, matrix-calculus appendix](#).

Trace: example

Noncommuting products with the same trace

Let

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 4 & -1 \\ 2 & 1 \end{bmatrix}.$$

Then

$$AB = \begin{bmatrix} 8 & 1 \\ 6 & 3 \end{bmatrix}, \quad BA = \begin{bmatrix} 4 & 5 \\ 2 & 7 \end{bmatrix}.$$

Thus $AB \neq BA$, while

$$\text{Tr}(AB) = 11 = \text{Tr}(BA).$$

Cyclic trace in OLS

Residual degrees of freedom in OLS

In $y = X\beta + u$, let $\text{rank}(X) = k$ and define

$$P_X = X(X^T X)^{-1} X^T, \quad M_X = I_n - P_X.$$

Cyclic step. Rotate the leading X to the end:

$$\text{Tr}(P_X) = \text{Tr}\left((X^T X)^{-1} X^T X\right) = \text{Tr}(I_k) = k,$$

$$\text{Tr}(M_X) = \text{Tr}(I_n) - \text{Tr}(P_X) = n - k.$$

Cyclic trace in OLS

Residual variance in OLS

Since $M_X X = 0$, $M_X^T = M_X$, and $M_X^2 = M_X$,

$$\hat{u} = M_X u, \quad \hat{u}^T \hat{u} = u^T M_X u = \text{Tr}(M_X u u^T).$$

If $E[uu^T | X] = \sigma^2 I_n$, then

$$E[\hat{u}^T \hat{u} | X] = \text{Tr}(M_X \sigma^2 I_n) = \sigma^2 (n - k).$$

Hence, if $n > k$,

$$\hat{\sigma}^2 = \frac{\hat{u}^T \hat{u}}{n - k}$$

is unbiased under $E[uu^T | X] = \sigma^2 I_n$.

Section 3

Determinants & inverses

Computing determinants

Determinants are only defined for square matrices. For a 2×2 matrix,

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc.$$

Example.

$$A = \begin{bmatrix} 3 & -2 \\ 4 & 1 \end{bmatrix}, \quad \det(A) = 3 \cdot 1 - (-2) \cdot 4 \\ = 3 - (-8) = 11.$$

A nonzero determinant will be a quick test for invertibility of a square matrix.

Video: [3Blue1Brown](#), *The determinant*.

2×2 determinants: quick practice

Compute each determinant by hand. Which matrices are singular?

1. $\det \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$

2. $\det \begin{bmatrix} -2 & 5 \\ 3 & 1 \end{bmatrix}$

3. $\det \begin{bmatrix} 6 & -3 \\ 2 & -1 \end{bmatrix}$

4. $\det \begin{bmatrix} 0 & 4 \\ -3 & 2 \end{bmatrix}$

5. $\det \begin{bmatrix} 1 & -0.8 \\ -0.5 & 1 \end{bmatrix}$

6. $\det \begin{bmatrix} 1 & a \\ b & 1 \end{bmatrix}$; when is it zero?

Target: the rule $ad - bc$ should become automatic before moving to larger determinants.

Six 2×2 determinants

$$(1) 4(3) - 1(2) = 10,$$

$$(2) (-2)(1) - 5(3) = -17,$$

$$(3) 6(-1) - (-3)(2) = 0.$$

Matrix (3) is singular.

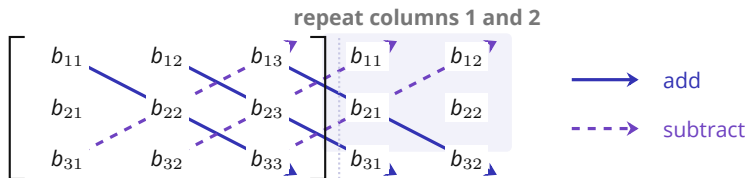
$$(4) 0(2) - 4(-3) = 12,$$

$$(5) 1 - (-0.8)(-0.5) = 0.6,$$

$$(6) 1 - ab.$$

Matrix (6) is singular when $ab = 1$.

Sarrus' rule



Formula. For a 3×3 matrix, Sarrus' rule gives

$$|B| = b_{11}b_{22}b_{33} + b_{12}b_{23}b_{31} + b_{13}b_{21}b_{32} \\ - b_{13}b_{22}b_{31} - b_{21}b_{12}b_{33} - b_{11}b_{23}b_{32}.$$

Range of the rule. Sarrus' rule computes the determinant of a 3×3 matrix. Cofactor expansion computes the determinant of a general $n \times n$ matrix.

Submatrices and minors

For $A \in \mathbb{R}^{m \times n}$, the submatrix A_{ij} is obtained by deleting row i and column j . If A is square, the determinant $\det(A_{ij})$ is the corresponding minor.

Construction. To form A_{12} , delete the first row and second column. To form A_{34} , delete the third row and fourth column. Preserve the order of all remaining rows and columns.

Cofactor expansion

Determinants of general $n \times n$ matrices

Expanding along any fixed row i , or symmetrically along any fixed column,

$$\det(A) = \sum_{j=1}^n (-1)^{i+j} a_{ij} \det(A_{ij}),$$

where the term $(-1)^{i+j} \det(A_{ij})$ is the cofactor associated with entry a_{ij} . For $A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$, the four cofactors are

$$C_{11} = 4, \quad C_{12} = -1, \quad C_{21} = -3, \quad C_{22} = 2, \quad \text{cof}(A) = \begin{bmatrix} 4 & -1 \\ -3 & 2 \end{bmatrix}.$$

Video: [Wrath of Math](#), *The cofactor definition of determinants (Laplace expansion)*.

Determinant properties

Products and transpose.

$$\det(AB) = \det(A) \det(B),$$
$$\det(A^T) = \det(A).$$

Invertibility test. A square matrix A is **invertible** when some square B satisfies $AB = BA = I$; that B is written A^{-1} . Then

$$\det(A) \neq 0 \quad \Leftrightarrow \quad A \text{ is invertible.}$$

Scaling. If $A \in \mathbb{R}^{n \times n}$, then

$$\det(cA) = c^n \det(A).$$

Singular case.

$$\det(A) = 0 \quad \Leftrightarrow \quad A \text{ is singular.}$$

Video: MIT 18.06, Gilbert Strang, *Properties of determinants*.

Determinant properties: quick practice

Let $A, B \in \mathbb{R}^{3 \times 3}$, with $\det(A) = -2$ and $\det(B) = 5$.

1. Find $\det(AB)$.

2. Find $\det(A^T)$.

3. Find $\det(3A)$.

4. Are A , B , and AB invertible?

Use the determinant properties directly.

Using determinant properties

Since $A, B \in \mathbb{R}^{3 \times 3}$, scaling all entries of A by 3 scales its determinant by 3^3 . Therefore

$$\det(AB) = \det(A) \det(B) = (-2)(5) = -10,$$

$$\det(A^T) = \det(A) = -2,$$

$$\det(3A) = 3^3 \det(A) = 27(-2) = -54.$$

Both given determinants are nonzero, so A and B are invertible. Their product is also invertible because $\det(AB) = -10 \neq 0$.

Matrix inversion

Two-by-two inverse. For $ad - bc \neq 0$,

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Computing the inverse. Swap the diagonal entries, change the off-diagonal signs, then divide by $ad - bc$. A square matrix is invertible if and only if its determinant is nonzero. For larger matrices, Gaussian elimination on the augmented matrix $[A \mid I]$ produces $[I \mid A^{-1}]$.

Video: 3Blue1Brown, *Inverse matrices, column space and null space*.

2×2 inverses: quick practice

For each matrix, first compute its determinant. If it is nonzero, find the inverse by hand.

1. $A_1 = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$

2. $A_2 = \begin{bmatrix} 3 & -1 \\ 2 & 1 \end{bmatrix}$

3. $A_3 = \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}$

2×2 inverses: quick practice

4. $A_4 = \begin{bmatrix} 0 & 2 \\ -1 & 3 \end{bmatrix}$

5. $A_5 = \begin{bmatrix} 1 & -0.8 \\ -0.5 & 1 \end{bmatrix}$

6. $A_6 = \begin{bmatrix} 1 & -a \\ -b & 1 \end{bmatrix}$; state the condition for A_6^{-1} to exist.

Habit: check $ad - bc \neq 0$, then swap the diagonal entries, change the off-diagonal signs, and divide by $ad - bc$.

Six 2×2 inverses

$$\det(A_1) = 1, \quad A_1^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix},$$

$$\det(A_2) = 5, \quad A_2^{-1} = \frac{1}{5} \begin{bmatrix} 1 & 1 \\ -2 & 3 \end{bmatrix},$$

$$\det(A_3) = 0, \quad A_3^{-1} \text{ does not exist.}$$

Six 2×2 inverses

$$\det(A_4) = 2, \quad A_4^{-1} = \frac{1}{2} \begin{bmatrix} 3 & -2 \\ 1 & 0 \end{bmatrix},$$

$$\det(A_5) = 0.6, \quad A_5^{-1} = \frac{1}{0.6} \begin{bmatrix} 1 & 0.8 \\ 0.5 & 1 \end{bmatrix},$$

$$\det(A_6) = 1 - ab, \quad A_6^{-1} = \frac{1}{1 - ab} \begin{bmatrix} 1 & a \\ b & 1 \end{bmatrix}, \quad ab \neq 1.$$

The adjugate formula

Inverses from cofactors

If $\det(A) \neq 0$, then

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A), \quad \operatorname{adj}(A) = \operatorname{cof}(A)^T,$$

where the cofactor matrix $\operatorname{cof}(A)$ has entries $[\operatorname{cof}(A)]_{ij} = (-1)^{i+j} \det(A_{ij})$.

$$\text{For } A = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}, \quad \det(A) = 2, \quad \operatorname{cof}(A) = \begin{bmatrix} 1 & 1 & -1 \\ 0 & 2 & 0 \\ 1 & -1 & 1 \end{bmatrix},$$

$$A^{-1} = \frac{1}{2} \operatorname{cof}(A)^T = \frac{1}{2} \begin{bmatrix} 1 & 0 & 1 \\ 1 & 2 & -1 \\ -1 & 0 & 1 \end{bmatrix}.$$

Video: [Khan Academy, Determinant and adjugate of a matrix.](#)

Income system and invertibility

From

$$Y = C + 50, \quad C = 100 + 0.8Y,$$

we obtain

$$A \begin{bmatrix} Y \\ C \end{bmatrix} = \begin{bmatrix} 50 \\ 100 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & -1 \\ -0.8 & 1 \end{bmatrix}.$$

Since

$$\det(A) = 1 - 0.8 = 0.2 \neq 0,$$

the matrix A is invertible.

Income determined by the inverse

$$A^{-1} = \frac{1}{0.2} \begin{bmatrix} 1 & 1 \\ 0.8 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 5 \\ 4 & 5 \end{bmatrix}.$$

Therefore

$$\begin{bmatrix} Y \\ C \end{bmatrix} = A^{-1} \begin{bmatrix} 50 \\ 100 \end{bmatrix} = \begin{bmatrix} 750 \\ 700 \end{bmatrix}.$$

Thus $Y = 750$ and $C = 700$.

Check: $750 = 700 + 50$ and $700 = 100 + 0.8(750)$.

Inverse properties

Basic identities. When it exists, the inverse is unique and

$$AA^{-1} = A^{-1}A = I, \quad (A^{-1})^{-1} = A.$$

Transpose and products.

$$(A^{-1})^T = (A^T)^{-1}, \quad (AB)^{-1} = B^{-1}A^{-1}.$$

The product rule requires A and B to be invertible and compatible.

The order reversal in $(AB)^{-1} = B^{-1}A^{-1}$ mirrors the transpose rule $(AB)^T = B^TA^T$.

Video: [Professor Dave Explains, Inverse matrices and their properties.](#)

Products and inverses

1. CONFORMABILITY AND MULTIPLICATION

Let

$$A = \begin{bmatrix} 4 & -1 \\ 6 & 9 \\ 2 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 3 \\ 3 & -2 \end{bmatrix}.$$

- (a) Is AB defined? If so, compute it.
- (b) Is BA defined? Explain using dimensions.

2. THREE INVERSES

Find the inverse of each matrix when it exists:

$$C_1 = \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}, \quad C_2 = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix},$$

$$C_3 = \begin{bmatrix} 3 & 0 \\ 1 & 5 \end{bmatrix}.$$

Verify both $C_1 C_1^{-1} = I$ and $C_1^{-1} C_1 = I$.

Products and inverses

The dimensions give $AB : (3 \times 2)(2 \times 2) \rightarrow 3 \times 2$, while BA is undefined because its inner dimensions are 2 and 3. Direct calculation gives

$$AB = \begin{bmatrix} -3 & 14 \\ 27 & 0 \\ 9 & 0 \end{bmatrix}.$$

The determinants of C_1, C_2, C_3 are 10, 0, 15, respectively. Hence

$$C_1^{-1} = \frac{1}{10} \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix}, \quad C_2^{-1} \text{ does not exist}, \quad C_3^{-1} = \frac{1}{15} \begin{bmatrix} 5 & 0 \\ -1 & 3 \end{bmatrix}.$$

$$C_1 C_1^{-1} = C_1^{-1} C_1 = \frac{1}{10} \begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix} = I_2.$$

A composite coefficient matrix

Direct and indirect coefficients

A two-sector model combines a direct coefficient matrix with an indirect channel:

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & 1 \\ 1 & 2 \end{bmatrix}.$$

- (a) Compute $M = A + BC$. Decide whether M^{-1} exists and find it when it does.
- (b) Replace A by $\tilde{A} = \begin{bmatrix} 2 & 2 \\ 0 & -3 \end{bmatrix}$. Decide whether $(\tilde{A} + BC)^{-1}$ exists.

A composite coefficient matrix

Multiplication, addition, and invertibility

First compute the shared product:

$$BC = \begin{bmatrix} -1 & -1 \\ 1 & 4 \end{bmatrix}.$$

Hence

$$M = A + BC = \begin{bmatrix} 1 & 0 \\ 2 & 4 \end{bmatrix}.$$

Since $\det(M) = 4$,

$$M^{-1} = \frac{1}{4} \begin{bmatrix} 4 & 0 \\ -2 & 1 \end{bmatrix}.$$

For the second calibration,

$$\tilde{A} + BC = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad \det(\tilde{A} + BC) = 0.$$

Therefore $(\tilde{A} + BC)^{-1}$ does not exist.

Eigenvalues and eigenvectors

Definition. A scalar λ is an eigenvalue of a square matrix A if

$$Av = \lambda v \quad \text{for some } v \neq 0,$$

and each such nonzero v is an eigenvector associated with λ .

Characteristic equation. Since $(A - \lambda I)v = 0$ has a nonzero solution, $A - \lambda I$ is singular:

$$\det(A - \lambda I) = 0.$$

In this course we compute eigenvalues for symmetric matrices; every eigenvalue of a symmetric matrix is real. For $A = \text{diag}(2, -1)$, the eigenvalues are 2 and -1 . The eigenvalues include 0 if and only if $\det(A) = 0$, that is, when $\text{rank}(A) < n$.

Video: [3Blue1Brown, *Eigenvectors and eigenvalues*](#). Further reading: [Boyd and Vandenberghe, slides for *Convex Optimization*](#).

Eigenvalues of a 2×2 matrix

Characteristic quadratic. For a symmetric 2×2 matrix,

$$A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}, \quad \det(A - \lambda I) = \lambda^2 - (a + c)\lambda + (ac - b^2) = 0.$$

The discriminant is $(a - c)^2 + 4b^2 \geq 0$, so both roots are real; substituting each root into $(A - \lambda I)v = 0$ gives the eigenvectors.

Example. For the symmetric matrix

$$S = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}, \quad \det(S - \lambda I) = \lambda^2 - 6\lambda + 8 = (\lambda - 2)(\lambda - 4) = 0.$$

The eigenvalue 4 has eigenvector $(1, 1)^T$ and the eigenvalue 2 has eigenvector $(1, -1)^T$.

Section 4

Rank & elimination

Elementary row operations & row echelon form

Operations.

1. interchange two rows;
2. multiply a row by a nonzero scalar;
3. add a multiple of one row to another row.

On $[A \mid b]$, these operations preserve the solution set of $Ax = b$.

What is preserved. Applied to the coefficient matrix A , elementary row operations are reversible. Hence they preserve

$\text{rank}(A)$ and $\ker(A)$.

Row echelon form. Zero rows are below nonzero rows; each leading entry lies strictly to the right of the leading entry above it; and all entries below a leading entry are zero. These leading entries are the **pivots**.

Video: [Professor Dave Explains, Elementary row operations and Gauss-Jordan elimination.](#)

Which matrices are in row echelon form

In row echelon form.

$$\left[\begin{array}{cc|c} 1 & -1 & 50 \\ 0 & 0.2 & 140 \end{array} \right]$$

the income and consumption example after $R_2 \leftarrow R_2 + 0.8R_1$

$$\left[\begin{array}{cccc} 2 & -1 & 0 & 4 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

column 2 carries no pivot; the zero row is at the bottom

$$\left[\begin{array}{ccc} 0 & 1 & 7 \\ 0 & 0 & 2 \end{array} \right]$$

the staircase may begin after a zero column

Outside row echelon form.

$$\left[\begin{array}{ccc} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 4 & 5 \end{array} \right]$$

a zero row sits above a nonzero row

$$\left[\begin{array}{ccc} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 0 & 0 & 0 \end{array} \right]$$

two leading entries share column 1

$$\left[\begin{array}{ccc} 1 & 2 & 3 \\ 0 & 0 & 4 \\ 0 & 5 & 6 \end{array} \right]$$

the last leading entry sits left of the one above

Each of the three is one row operation from echelon form. Interchanging the last two rows repairs the first and the third; $R_2 \leftarrow R_2 - 2R_1$ repairs the second.

Solving a three-variable system

System. Solve for x_1, x_2, x_3 :

$$x_1 + x_2 + x_3 = 4,$$

$$2x_1 + x_2 + x_3 = 5,$$

$$x_1 + 2x_2 + 3x_3 = 9.$$

Row reduction. $R_2 \leftarrow R_2 - 2R_1, \quad R_3 \leftarrow R_3 - R_1$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 2 & 1 & 1 & 5 \\ 1 & 2 & 3 & 9 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & -1 & -1 & -3 \\ 0 & 1 & 2 & 5 \end{array} \right]$$

$$R_3 \leftarrow R_3 + R_2 \quad \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & -1 & -1 & -3 \\ 0 & 0 & 1 & 2 \end{array} \right]$$

Back-substitution gives

$$x_3 = 2, \quad -x_2 - x_3 = -3 \Rightarrow x_2 = 1, \quad x_1 + x_2 + x_3 = 4 \Rightarrow x_1 = 1.$$

Thus $(x_1, x_2, x_3) = (1, 1, 2)$. The three pivots give rank 3, so the system has a unique solution.

Video: MIT 18.06, Gilbert Strang, *Elimination with matrices*.

Determinant from elimination

What each operation does. Adding a multiple of one row to another leaves $\det$ unchanged; interchanging two rows reverses its sign; multiplying a row by c multiplies it by c . So for a square A carried to an echelon form U by r interchanges and no scaling,

$$\det A = (-1)^r u_{11} u_{22} \cdots u_{nn}.$$

The system on the previous slide. That elimination used no interchange and no scaling, and its pivots are $1, -1, 1$, so

$$\det \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} = 1 \cdot (-1) \cdot 1 = -1.$$

Cofactor expansion along the first row agrees: $1(3 - 2) - 1(6 - 1) + 1(4 - 1) = -1$.

When A is invertible its n pivots sit on the diagonal of U , so the determinant is the product of the pivots. A zero on that diagonal leaves fewer than n pivots and a determinant of 0.

Computing rank

Count pivots. If a row echelon form of A has r pivots, then

$$\text{rank}(A) = r.$$

Free variables. If $A \in \mathbb{R}^{m \times n}$, then

$$\# \text{ non-pivot columns} = n - r,$$

$$\# \text{ free variables in } Ax = 0 = n - r.$$

Independent columns. The pivot positions identify the columns carrying independent information. The corresponding *original* columns of A form a basis of $\text{im}(A)$.

Rank-nullity in computation.

$$n = r + \dim \ker(A), \quad \dim \ker(A) = n - r.$$

Video: Wrath of Math, *Finding the rank of a 3x3 matrix from row echelon form.*

Rank, columns, and the kernel

Let $A = [a_1 \cdots a_n] \in \mathbb{R}^{m \times n}$ and $r = \text{rank}(A)$.

Columns.

$$Ax = 0 \Leftrightarrow x_1 a_1 + \cdots + x_n a_n = 0.$$

Independence.

$$\begin{aligned} a_1, \dots, a_n \text{ independent} &\Leftrightarrow \ker(A) = \{0\} \\ &\Leftrightarrow \text{rank}(A) = n. \end{aligned}$$

Row reduction.

$$\begin{aligned} r \text{ pivots} &\Leftrightarrow \text{rank}(A) = r, \\ n - r \text{ free variables} &\Leftrightarrow \dim \ker(A) = n - r. \end{aligned}$$

Rank and nullity.

$$\text{rank}(A) + \dim \ker(A) = n.$$

One elimination delivers both numbers: r pivot columns and $n - r$ free variables.

Rank examples

Rank one. The two rows of

$$A = \begin{bmatrix} 2 & 0 \\ 2 & 0 \end{bmatrix}$$

are equal, so A has one pivot and rank 1.

Rank two. Row reduction of

$$B = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 3 & 4 \end{bmatrix}$$

gives two pivots, so B has rank 2.

Computational test. Count the nonzero rows of a row echelon form. One step $R_2 \leftarrow R_2 - R_1$ sends A to $\begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$, so $\text{rank}(A) = 1$ and $Ax = 0$ has $2 - 1 = 1$ free variable; B already has two nonzero rows, so $\text{rank}(B) = 2$ and $Bx = 0$ has $3 - 2 = 1$ free variable. The pivot columns are the ones that pin down a variable, and the remaining $n - r$ columns carry the free variables.

OLS full rank

Full column rank in OLS

REGRESSION SETUP

In the linear regression model $y = X\beta + u$
with $X \in \mathbb{R}^{n \times k}$, full column rank means

$$\text{rank}(X) = k.$$

Ordinary least squares uses

$$\hat{\beta} = (X^T X)^{-1} X^T y.$$

IDENTIFICATION MEANING

Full column rank is equivalent to $X^T X$ being invertible.

If one regressor is an exact linear combination of the others, the coefficients are not separately identified.

Example: income in euros and income in cents are the same variable in different units. Including both creates a redundant column.

Section 5

Linear systems

General form & outcomes

For $A \in \mathbb{R}^{m \times n}$, consider $Ax = b$, and let $[A \mid b]$ be the augmented matrix.

Existence.

$$\begin{aligned} Ax = b \text{ solvable} &\Leftrightarrow b \in \text{im}(A) \\ &\Leftrightarrow \text{rank}(A) = \text{rank}([A \mid b]). \end{aligned}$$

If $\text{rank}(A) < \text{rank}([A \mid b])$, there is **no solution**.

Multiplicity. If the system is consistent and $r = \text{rank}(A)$, then

$$\begin{aligned} \{x : Ax = b\} &= x_p + \ker(A), \\ \dim \ker(A) &= n - r. \end{aligned}$$

Thus the solution is **unique** when $r = n$, and there are **infinitely many** solutions when $r < n$.

Video: MIT 18.06, Gilbert Strang, *Solving $Ax=b$: Row Reduced Form R*. Video: Mulkek, *A unique solution, no solution, or infinitely many solutions*.

Two equations, two unknowns

System.

$$ax_1 + bx_2 = u, \quad cx_1 + dx_2 = v, \quad A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad [A \mid r] = \left[\begin{array}{cc|c} a & b & u \\ c & d & v \end{array} \right].$$

If $\det A = ad - bc \neq 0$. There is a unique solution:

$$x_1 = \frac{du - bv}{ad - bc}, \quad x_2 = \frac{av - cu}{ad - bc}.$$

If $\det A = 0$. Use ranks. Equal ranks below 2 give infinitely many solutions; a larger augmented rank gives no solution.

Equivalent tests for invertibility

For a square matrix $A \in \mathbb{R}^{n \times n}$, the following conditions are equivalent.

Row reduction.

A invertible $\Leftrightarrow \text{rank}(A) = n$
 $\Leftrightarrow n$ pivots.

Linear system. For every $b \in \mathbb{R}^n$, $Ax = b$
has the unique solution

$$x = A^{-1}b.$$

Column space.

$\ker(A) = \{0\} \Leftrightarrow$ columns independent
 $\Leftrightarrow \text{im}(A) = \mathbb{R}^n$.

Determinant.

$$\det(A) \neq 0.$$

Video: [James Hamblin, The invertible matrix theorem.](#)

A Leontief economy

Input-output equation

Gross output covers intermediate demand and final demand:

$$x = Ax + d \quad \Leftrightarrow \quad (I - A)x = d.$$

Here

$$d = \begin{bmatrix} 60 \\ 40 \\ 50 \end{bmatrix}.$$

Coefficient matrix and solution

$$I - A = \begin{bmatrix} .8 & -.1 & -.1 \\ -.3 & .8 & -.1 \\ -.1 & -.2 & .8 \end{bmatrix}$$

$$\det(I - A) = 0.457 \neq 0.$$

Therefore

$$x = (I - A)^{-1}d = \begin{bmatrix} 100 \\ 100 \\ 100 \end{bmatrix}.$$

The computed output vector is nonnegative and can be checked by substituting it into $(I - A)x = d$.

Video: [Justin Eloriaga, *The Leontief input-output model*](#).

Inconsistent and underidentified systems

Incompatible targets

$\text{rank}(A) < \text{rank}([A \mid b])$:

$$q_1 + q_2 + q_3 = 1,$$

$$2q_1 + 2q_2 + 2q_3 = 2,$$

$$q_1 + q_2 + q_3 = 3.$$

The same aggregate requirement is set equal to two different values.

Relative prices without normalization

$\text{rank}(A) = \text{rank}([A \mid 0]) = 2 < 3$:

$$p_1 - p_2 = 0,$$

$$p_2 - p_3 = 0,$$

$$2p_1 - p_2 - p_3 = 0.$$

The solution is $p = t(1, 1, 1)$, so one scale degree of freedom remains.

Same rank language, two different failures: inconsistency means no point satisfies all equations; underidentification means a whole affine set remains.

Video: [Mario's Math Tutoring, Consistent independent, dependent, and inconsistent systems.](#)

Walras' law and price normalization

Redundant clearing equation and price normalization

Let $z(p) \in \mathbb{R}^L$ be aggregate excess demand at a strictly positive price vector p . Suppose excess demand is homogeneous of degree zero and satisfies Walras' law:

$$z(tp) = z(p) \quad (t > 0), \quad p^\top z(p) = 0.$$

If $z_1(p) = \dots = z_{L-1}(p) = 0$, then Walras' law and $p_L > 0$ imply $z_L(p) = 0$. Thus the last market-clearing equation is redundant once the first $L - 1$ hold. Homogeneity implies that p^* and tp^* produce the same excess demand, so the normalization

$$\sum_{\ell=1}^L p_\ell = 1 \quad \Rightarrow \quad \hat{p} = \frac{p^*}{\sum_{\ell=1}^L p_\ell^*}$$

selects one representative of the equilibrium price ray.